

KCPC Week 1 Reading List

Reasoning Foundations: Logic, Invariants & Heuristics

How to Solve It George Pólya (Princeton University Press, 2004)

A classic introduction to solving unfamiliar problems. Pólya's four-stage method—understand the problem, devise a plan, carry it out, and look back—provides a useful routine for every puzzle in Week 1. The book develops practical heuristics through approachable mathematical examples rather than presenting a catalogue of formulas.

The Art and Craft of Problem Solving Paul Zeitz (Wiley, 2007)

A lively guide to the habits behind successful problem solving. Zeitz explores experimentation, pattern recognition, symmetry, parity, invariants, contradiction, and extremal arguments. It is especially useful for learning how to make progress when a problem gives no obvious starting point, and includes many exercises suitable for ambitious beginners.

Problem-Solving Strategies Arthur Engel (Springer, 1998)

A technique-centred collection for readers who want to develop beyond isolated tricks. Engel gives substantial attention to invariants, colouring, induction, extremal arguments, and the pigeonhole principle. The large bank of worked examples makes it particularly relevant to the proof-based puzzles and progressively harder exercises in Week 1.

Mathematical Puzzles: A Connoisseur's Collection Peter Winkler (A K Peters, 2004)

An engaging selection of elegant puzzles from combinatorics, probability, geometry, and logic. The problems are short to state but often require exactly the kind of decisive insight practised in Week 1. Winkler's explanations are concise and reveal why the central idea works without hiding the creative leap.

The Moscow Puzzles Boris A. Kordemsky (Dover Publications, 1992)

A classic collection of recreational problems involving logic, parity, counting, arrangements, and geometric constructions. The puzzles range from quick observations to challenges requiring sustained reasoning. Its informal style makes it easy to browse, while the variety encourages readers to decide for themselves which Week 1 technique might apply.

Proofs from THE BOOK Martin Aigner and Günter M. Ziegler (Springer, 2018)

A collection of exceptionally elegant proofs from across mathematics. The chapters demonstrate how parity, counting, induction, contradiction, and carefully chosen representations can produce arguments that are both rigorous and memorable. Not every chapter is introductory, but many examples can be enjoyed with school-level mathematics and curiosity.